

Edexcel IAL Physics A-level

Topic 5.6: Astrophysics and Cosmology

Topics Covered

- 5.6.154–5.6.155 Gravitational Fields and Field Strength
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Note on formulas: All mathematical relationships in this document are written in plain English text (e.g. " L equals sigma times A times T to the power 4") to support text-based retrieval systems.

1. Gravitational Fields

Spec ref: 5.6.154, 5.6.155

Force Fields and Gravitational Fields

A **force field** is a region of space in which an object experiences a non-contact force. Force fields are represented as vectors (showing direction) and by **field-line diagrams** where the spacing between lines indicates field strength: closer lines mean stronger force.

A **gravitational field** is a force field in which any object that has mass experiences an attractive gravitational force. Every mass creates its own gravitational field. Gravity is **always attractive** — it never repels.

Types of Gravitational Field

There are two types of gravitational field:

- **Uniform field** — the gravitational force is the same at every point, shown by parallel, equally-spaced field lines all pointing in the same direction. Near the Earth's surface the field is treated as approximately uniform.
- **Radial field** — field lines point towards the central mass from all directions (like spokes on a wheel). As distance from the centre increases, the lines spread apart, indicating a weaker force. The Earth's gravitational field is radial on a large scale.

Gravitational Field Strength (g)

Gravitational field strength g is defined as the gravitational force per unit mass at a point. Its unit is **newtons per kilogram (N per kg)**, equivalent to m per s squared.

$$g = F \div m$$

where g is gravitational field strength (N per kg), F is the gravitational force (N), m is the mass of the object (kg)

In a uniform field, g is constant everywhere. In a radial field, g varies with position. At the Earth's surface, $g \approx 9.81$ N per kg.

The **weight** of an object is the gravitational force on it:

$$\text{Weight} = m \times g$$

where m is mass (kg) and g is gravitational field strength (N per kg). Weight is measured in newtons.

Worked Example — Weight on the Moon

What force does a 2000 kg elephant experience on the Moon's surface? The Moon's surface gravitational field strength is 1.62 N per kg.

Solution:

Using Weight equals m times g:

$$F = 2000 \text{ kg} \times 1.62 \text{ N per kg}$$

$$F = 3240 \text{ N}$$

The elephant would weigh 3240 N on the Moon, compared to about 19 620 N on Earth.

Key Terms

Force field	A region of space in which an object experiences a non-contact force.
Field lines	Lines used to represent a field; spacing indicates field strength.
Uniform field	A field in which the force is the same at every point.
Radial field	A field whose field lines radiate outward from (or inward to) a point.
g (field strength)	Gravitational force per unit mass; $g = F/m$; units N per kg.

Exam Practice Questions

Q1. Define gravitational field strength and state its SI unit. [2 marks]

Answer: Gravitational field strength is the gravitational force per unit mass experienced at a point in the field ($g = F/m$). Its SI unit is newtons per kilogram (N per kg), equivalent to m per s squared .

Q2. Explain the difference between a uniform and a radial gravitational field, giving one example of each. [3 marks]

Answer: Uniform field: the gravitational force is the same magnitude and direction at every point; shown by parallel, equally-spaced field lines. Example: near the Earth's surface. Radial field: the force varies with position; field lines point inward toward the source mass and become more spread apart with distance, indicating decreasing field strength. Example: the gravitational field of a planet or star at large distances.

Q3. An astronaut of mass 80 kg stands on the surface of Mars, where the gravitational field strength is 3.72 N per kg. Calculate the astronaut's weight on Mars. [2 marks]

Answer: Weight = $m \times g = 80 \text{ kg} \times 3.72 \text{ N per kg} = 297.6 \text{ N}$

2. Newton's Law of Universal Gravitation

Spec ref: 5.6.156

Statement of the Law

Gravity acts on all objects with mass and is **always attractive**. Newton's law of gravitation states:

"The gravitational force between two point masses is directly proportional to the product of their masses and inversely proportional to the square of the distance between their centres."

$$F = G m_1 m_2 \div r^2$$

where G is the gravitational constant $= 6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$, m_1 and m_2 are the masses (kg), r is the centre-to-centre distance (m), and F is the gravitational force (N).

Inverse-square law: doubling the separation r reduces the force F to one quarter of its value. By Newton's third law, both masses experience equal and opposite forces.

The gravitational constant G is universal — it applies everywhere in the Universe. It was first measured by Henry Cavendish in 1798 using a torsion balance experiment.

Weighing the Earth

Newton's law allows us to calculate planetary masses from orbital data. The Moon keeps the Earth in mind because gravity provides the centripetal force:

$$G m_E m_M \div r^2 = m_M v^2 \div r$$

Simplifying: $m_E = r v^2 / G$. Using the Moon's orbital radius $r = 3.84 \times 10^8 \text{ m}$ and period $T = 2.36 \times 10^6 \text{ s}$ gives $m_E \approx 6.01 \times 10^{24} \text{ kg}$.

Worked Example — Force Between Two Neutrons

Calculate the gravitational force between two neutrons in the Crab Nebula that are 2 m apart. Mass of a neutron $= 1.67 \times 10^{-27} \text{ kg}$.

Solution:

$$F = G m_1 m_2 / r^2$$

$$F = (6.67 \times 10^{-11} \times 1.67 \times 10^{-27} \times 1.67 \times 10^{-27}) / 2^2$$

$$F = 4.65 \times 10^{-65} \text{ N}$$

This tiny force explains why gravity is the weakest of the four fundamental forces.

Worked Example — Gravitational Force Earth–Moon

Calculate the gravitational force between the Earth and the Moon. Mass of Earth = 6.00×10^{24} kg, mass of Moon = 7.35×10^{22} kg, orbital radius = 3.84×10^8 m.

Solution:

$$F = G m_E m_M / r^2$$

$$F = (6.67 \times 10^{-11} \times 6.00 \times 10^{24} \times 7.35 \times 10^{22}) / (3.84 \times 10^8)^2$$

$$F = 1.99 \times 10^{20} \text{ N}$$

This enormous force maintains the Moon in its orbit around the Earth.

Key Terms

Gravitational constant G	$G = 6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$; universal constant.
Inverse-square law	Force is inversely proportional to the square of the separation: $F \propto 1/r^2$.
Centre-to-centre distance	r is always measured between the centres of mass of the two objects.

Exam Practice Questions

Q1. State Newton's law of universal gravitation. [2 marks]

Answer: The gravitational force between two point masses is directly proportional to the product of their masses and inversely proportional to the square of the distance between their centres: $F = Gm_1m_2/r^2$.

Q2. Calculate the gravitational force between the Earth (mass 6.0×10^{24} kg) and the Sun (mass 2.0×10^{30} kg), given the mean Earth–Sun distance is 1.5×10^{11} m. [3 marks]

Answer: $F = G m_E m_S / r^2 = (6.67 \times 10^{-11} \times 6.0 \times 10^{24} \times 2.0 \times 10^{30}) / (1.5 \times 10^{11})^2 = 3.56 \times 10^{22} \text{ N}$

Q3. Explain why the inverse-square nature of gravity means gravitational attraction cannot be shielded from, unlike electric force. [3 marks]

Answer: The gravitational force falls off as $1/r^2$ but never reaches zero at any finite distance. Gravity acts on all mass and there is no negative mass to cancel it out (unlike electric charges where positive and negative can cancel). Therefore there is no way to screen or shield a mass from the gravitational field of another mass.

3. Gravitational Field Strength in a Radial Field

Spec ref: 5.6.157

Deriving g for a Radial Field

For a radial field produced by a point mass (or spherically symmetric body), the gravitational field strength g **varies with distance r** from the centre. We derive the formula by combining Newton's law of gravitation with the definition of g :

Step 1: Gravitational force on mass m placed in field of mass M :

$$F = G M m \div r^2$$

Step 2: Definition of gravitational field strength: $g = F / m$, so $F = mg$.

Step 3: Set equal and cancel m :

$$g = G M \div r^2$$

where $G = 6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$, M is the mass of the body causing the field (kg), r is the distance from the centre of that body (m), and g is the gravitational field strength (N per kg).

Inverse-square law: $g \propto 1/r^2$. Doubling the distance from the centre reduces g to one quarter of its original value.

Field strength is independent of the test mass. The formula for g contains only M (the source mass) and r — not the mass of the object placed in the field.

Worked Example — g at a Geostationary Orbit

Calculate the gravitational field strength caused by the Earth at the orbit of a geostationary satellite, which is at height $h = 35\,800 \text{ km}$ above the Earth's surface. Radius of Earth $R_E = 6400 \text{ km}$; mass of Earth $= 6.01 \times 10^{24} \text{ kg}$.

Solution:

Distance from Earth's centre: $r = R_E + h = 6400 + 35800 = 42200 \text{ km} = 4.22 \times 10^7 \text{ m}$

$g = G M / r^2$

$$g = (6.67 \times 10^{-11} \times 6.01 \times 10^{24}) / (4.22 \times 10^7)^2$$

$$g = 0.23 \text{ N per kg}$$

This is much smaller than at the Earth's surface (9.81 N per kg), as expected.

Worked Example — g at the Earth's Surface

Verify that the gravitational field strength at the Earth's surface is approximately 9.81 N per kg. Mass of Earth $= 5.97 \times 10^{24} \text{ kg}$; radius of Earth $= 6.38 \times 10^6 \text{ m}$.

Solution:

$g = G M / r^2$

$$g = (6.67 \times 10^{-11} \times 5.97 \times 10^{24}) / (6.38 \times 10^6)^2$$

$$g = 9.77 \text{ N per kg} \approx 9.81 \text{ N per kg (slight difference due to Earth not being a perfect sphere).}$$

Exam Practice Questions

Q1. Derive an expression for the gravitational field strength g at a distance r from the centre of a mass M . [3 marks]

Answer: The gravitational force on a test mass m at distance r is $F = GMm/r^2$ (Newton's law). Gravitational field strength is defined as $g = F/m$. Substituting: $g = GMm / (r^2 \times m) = GM / r^2$.

Q2. Calculate the gravitational field strength at the orbit of the ISS, which orbits at approximately 400 km above the Earth's surface. Mass of Earth = 5.97×10^{24} kg; radius of Earth = 6.38×10^6 m. [3 marks]

Answer: $r = R_E + h = 6.38 \times 10^6 + 4.0 \times 10^5 = 6.78 \times 10^6$ m; $g = GM / r^2 = (6.67 \times 10^{-11} \times 5.97 \times 10^{24}) / (6.78 \times 10^6)^2 = 8.66$ N per kg. This is close to but less than surface g , which is why astronauts on the ISS experience near-weightlessness — they are in continuous free fall.

Q3. Explain why g is independent of the mass of the object placed in the field. [2 marks]

Answer: $g = GM/r^2$ contains only M (the source mass) and r (the distance). When we derive g by combining $F = GMm/r^2$ with $g = F/m$, the test mass m cancels out. Therefore the field strength at any point depends only on the source mass and the distance, not on the object experiencing the field.

4. Gravitational Potential

Spec ref: 5.6.158

Definition of Gravitational Potential

Gravitational potential V at a point is the **work done per unit mass** when moving an object from infinity to that point.

Key points:

- Gravitational potential at infinity is defined as **zero**.
- As an object moves from infinity into a gravitational field, energy is released, so gravitational potential is **always negative** (or zero at infinity).
- The more negative V is, the more tightly an object is bound to the source mass.

Formula for Gravitational Potential (Radial Field)

$$V = -G M \div r$$

where $G = 6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$, M = mass of the body creating the field (kg), r = distance from the centre of that body (m), V = gravitational potential (J per kg). The negative sign reflects that work must be done to move a mass away from a gravitating body.

Gravitational Potential Difference and Work Done

Gravitational potential difference ΔV is the energy per unit mass needed to move an object between two points in a gravitational field.

$$\text{Work done} = m \times \Delta V$$

where m is the mass of the object moved (kg) and ΔV is the change in gravitational potential (J per kg).

The relationship between gravitational field strength and potential is:

$$g = -dV \div dx$$

g is the negative gradient of the potential-distance graph. For small distances: $g \times \Delta x = -\Delta V$.

Worked Example — Gravitational Potential at Geostationary Orbit

Calculate the gravitational potential caused by the Earth at the orbit of a geostationary satellite. $r = 4.22 \times 10^7 \text{ m}$; $M = 6.02 \times 10^{24} \text{ kg}$.

Solution:

$$V = -G M / r$$

$$V = -(6.67 \times 10^{-11} \times 6.02 \times 10^{24}) / (4.22 \times 10^7)$$

$$V = -9.52 \times 10^6 \text{ J per kg} = -9.52 \text{ MJ per kg}$$

The negative sign confirms energy must be supplied to move a satellite from this orbit out to infinity.

Worked Example — Work Done Moving a Satellite

Calculate the minimum energy needed to move a 500 kg satellite from the Earth's surface ($V_{\text{surface}} = -62.6$ MJ per kg) to geostationary orbit ($V_{\text{orbit}} = -9.52$ MJ per kg).

Solution:

$$\Delta V = V_{\text{orbit}} - V_{\text{surface}} = -9.52 - (-62.6) = +53.08 \text{ MJ per kg}$$

$$\text{Work done} = m \times \Delta V = 500 \times 53.08 \times 10^6$$

$$\text{Work done} = 2.65 \times 10^{10} \text{ J} = 26.5 \text{ GJ}$$

This enormous energy requirement explains why rocket fuel must be so massive.

Exam Practice Questions

Q1. Define gravitational potential at a point in a field. [2 marks]

Answer: Gravitational potential at a point is the work done per unit mass when moving an object from infinity to that point. It is always negative (or zero at infinity) because gravitational fields are attractive and energy is released when an object moves inward.

Q2. Explain why gravitational potential is always negative. [2 marks]

Answer: Gravitational potential is defined as zero at infinity. As an object moves from infinity towards a mass, work is done by the gravitational field (energy is released), so the potential decreases below zero. Since the gravitational force is always attractive, moving away from a mass always requires energy input, meaning all finite distances have negative potential.

Q3. A satellite of mass 250 kg is moved from orbit A ($V = -30.0$ MJ per kg) to orbit B ($V = -12.0$ MJ per kg). Calculate the work done on the satellite. [2 marks]

Answer: $\Delta V = V_{\text{B}} - V_{\text{A}} = -12.0 - (-30.0) = +18.0$ MJ per kg. Work done = $m \times \Delta V = 250 \times 18.0 \times 10^6 = 4.50 \times 10^9$ J. Positive work done means energy must be supplied to move the satellite to the higher orbit.

5. Comparing Electric and Gravitational Fields

Spec ref: 5.6.159

Mathematical Similarities

Both gravitational and electric fields obey an inverse-square law and share the same mathematical form for their equations, with different physical quantities playing analogous roles:

Quantity	Gravitational Field	Electric Field
Force	$F = G m_1 m_2 / r^2$	$F = k q_1 q_2 / r^2$
Field Strength	$g = G M / r^2$	$E = k Q / r^2$
Potential	$V = -G M / r$	$V = k Q / r$
Source	Mass M	Charge Q
Acts on	Mass m	Charge q
Constant	$G = 6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$	$k = 9.0 \times 10^9 \text{ N m}^2 \text{ C}^{-2}$

Key Differences

Difference	Gravitational Field	Electric Field
Direction of force	Always attractive (never repulsive)	Can be attractive (opposite charges) or repulsive (like charges)
Shielding	Cannot be shielded — all masses are affected	Can be shielded using a Faraday cage
Relative strength	Extremely weak — weakest fundamental force	Much stronger than gravity (by ~40 orders of magnitude for electron-proton pair)
Potential sign	Always negative	Positive for positive charges, negative for negative charges

Exam Practice Questions

Q1. State two similarities and two differences between electric and gravitational fields. [4 marks]

Answer: Similarities: (1) Both follow an inverse-square law for force ($F \propto 1/r^2$). (2) Both can be represented by field lines and can be either uniform or radial. Differences: (1) Gravity is always attractive; electric force can be attractive or repulsive. (2) Gravity acts on mass; electric force acts on charge. (Accept also: gravity cannot be shielded; the electromagnetic force is much stronger.)

Q2. Compare the gravitational field strength of the Earth and the electric field strength of a proton at the same distance. Comment on the relative magnitudes. [4 marks]

Answer: For a proton ($q = 1.6 \times 10^{-19} \text{ C}$, $m = 1.67 \times 10^{-27} \text{ kg}$) at distance r : Gravitational $g = Gm / r^2$; Electric $E = kq / r^2$. Ratio $E/g = kq / (Gm) = (9.0 \times 10^9 \times 1.6 \times 10^{-19}) / (6.67 \times 10^{-11} \times 1.67 \times 10^{-27}) \approx 1.3 \times 10^{29}$. The electric field strength is about 10^{29} times greater than the gravitational field strength, confirming that gravity is by far the weakest of the fundamental forces.

6. Orbital Motion — Kepler's Third Law

Spec ref: 5.6.160

Gravitational Force as Centripetal Force

For an object in circular orbit, the gravitational force **provides the centripetal force**. Setting the two expressions equal:

$$m v^2 / r = G M m / r^2$$

Left side: centripetal force (N). Right side: gravitational force (N). m = mass of orbiting object (kg), v = orbital speed (m/s), r = orbital radius (m), M = mass of central body (kg).

Cancelling m and one factor of r gives the orbital speed:

$$v^2 = G M / r$$

or equivalently: $v = \sqrt{G M / r}$. Orbital speed decreases as orbital radius increases.

Deriving Kepler's Third Law

For a circular orbit, the orbital speed is related to the period T and radius r by:

$$v = 2 \pi r / T$$

Squaring and substituting into $v^2 = GM/r$:

$$4 \pi^2 r^2 / T^2 = G M / r$$

Rearranging to make T^2 the subject:

$$T^2 = (4 \pi^2 / G M) \times r^3$$

where T is the orbital period (s), r is the orbital radius (m), M is the mass of the central body (kg), G is the gravitational constant. Since $4\pi^2/GM$ is constant for a given central body, $T^2 \propto r^3$ (Kepler's Third Law).

Kepler's Third Law: The square of the orbital period is directly proportional to the cube of the orbital radius. This applies to all satellites orbiting the same central body.

Worked Example — Geostationary Satellite Orbital Radius

A geostationary satellite has a period $T = 24$ hours = 86 400 s. Find its orbital radius. Mass of Earth = 6.0×10^{24} kg.

Solution:

$$T^2 = (4\pi^2 / GM) \times r^3$$

$$r^3 = G M T^2 / 4 \pi^2$$

$$r^3 = (6.67 \times 10^{-11} \times 6.0 \times 10^{24} \times 86400^2) / (4 \times \pi^2)$$

$$r^3 = 7.54 \times 10^{22} \text{ m}^3$$

$$r = (7.54 \times 10^{22})^{1/3} = 4.22 \times 10^7 \text{ m} = 42\,200 \text{ km}$$

This confirms the geostationary orbit height of about 35 800 km above the Earth's surface.

Worked Example — Mass of the Sun from Earth's Orbit

Earth's orbital radius $r = 1.5 \times 10^{11}$ m, orbital period $T = 3.15 \times 10^7$ s. Calculate the mass of the Sun.

Solution:

From $T^2 = (4\pi^2 / GM) \times r^3$:

$$M = 4\pi^2 r^3 / (GT^2)$$

$$M = (4\pi^2 \times (1.5 \times 10^{11})^3) / (6.67 \times 10^{-11} \times (3.15 \times 10^7)^2)$$

$$M = 2.0 \times 10^{30} \text{ kg}$$

Exam Practice Questions

Q1. Derive Kepler's Third Law starting from the equation for gravitational force. [5 marks]

Answer: Step 1: Equate gravitational force and centripetal force: $GMm/r^2 = mv^2/r$. Step 2: Cancel m and r to get $v^2 = GM/r$. Step 3: For circular orbit, $v = 2\pi r/T$ so $v^2 = 4\pi^2 r^2 / T^2$. Step 4: Substitute: $4\pi^2 r^2 / T^2 = GM/r$. Step 5: Rearrange: $T^2 = (4\pi^2 / GM) \times r^3$. Since $4\pi^2 / GM$ is constant, $T^2 \propto r^3$. QED.

Q2. Jupiter has four large moons (Galilean moons). Io orbits at $r = 4.22 \times 10^8$ m with period $T = 1.53 \times 10^5$ s. Ganymede orbits at $r = 1.07 \times 10^9$ m. Use Kepler's Third Law to find Ganymede's orbital period. [3 marks]

Answer: $T^2 / r^3 = \text{constant}$. So $T_G^2 / r_G^3 = T_I^2 / r_I^3$.
 $T_G^2 = T_I^2 \times (r_G / r_I)^3 = (1.53 \times 10^5)^2 \times (1.07 \times 10^9 / 4.22 \times 10^8)^3 = 2.34 \times 10^{10} \times 16.26 = 3.81 \times 10^{11}$.
 $T_G = 6.17 \times 10^5$ s (approximately 7.1 days).

7. Black Body Radiators and Radiation Curves

Spec ref: 5.6.161

Black Body Radiators

A **black body radiator** is a theoretical object that is a **perfect emitter and absorber of all wavelengths of electromagnetic radiation**. It emits EM radiation at every wavelength; the amount at each wavelength depends only on the object's temperature. A real example is a cavity (box) with a tiny hole: radiation entering the hole bounces around inside and is absorbed with near-perfect efficiency.

Stars behave approximately as black bodies, which allows us to determine their temperatures from the spectrum of light they emit.

Radiation Curves

A **radiation curve** (or black-body spectrum) is a graph of intensity (power per unit area per unit wavelength) against wavelength for a black body at a fixed temperature.

Key features of radiation curves:

- The curve starts at zero for very short wavelengths, rises to a **peak at λ_{max}** (peak wavelength), then gradually falls at longer wavelengths.
- As temperature increases, the **peak shifts to shorter wavelengths** — Wien's displacement law.
- As temperature increases, the **total area under the curve increases greatly** — Stefan-Boltzmann law: the total power radiated increases as T to the power 4.
- A hotter black body emits more radiation at every wavelength compared to a cooler one.

Connecting Temperature to Observed Star Colour

Temperature	Peak λ_{max}	Observed Colour
~2000–3500 K	Infrared–red, > 700 nm	Red (cool stars: M class)
~5800 K (Sun)	~500 nm (yellow-green)	Yellow-white (G class)
~10 000 K	~290 nm (UV)	Blue-white (A/B class)
~25 000 K	~115 nm (far UV)	Blue (O class)

Exam Practice Questions

Q1. Define a black body radiator. [2 marks]

Answer: A black body radiator is a theoretical object that is a perfect emitter and absorber of all possible wavelengths of electromagnetic radiation. No radiation is reflected. The spectrum of radiation it emits depends only on its temperature.

Q2. Describe how the black body radiation curve changes as the temperature of the black body increases. [3 marks]

Answer: (1) The peak wavelength λ_{max} shifts to shorter wavelengths (higher frequency). (2) The peak intensity increases (the maximum of the curve rises). (3) The total area under the curve increases, meaning the total power emitted per unit area increases — proportional to T^4 (Stefan-Boltzmann law). (4) The curve shifts to higher intensities at every wavelength.

Q3. Explain why a star with a surface temperature of 3000 K appears red. [2 marks]

Answer: At 3000 K the peak wavelength is in the infrared region. Only the long-wavelength end of the visible spectrum (red light, ~700 nm) is emitted significantly. Therefore the star appears predominantly red when observed from Earth.

8. The Stefan-Boltzmann Law

Spec ref: 5.6.162

Statement of the Law

The **Stefan-Boltzmann law** relates the total power output (luminosity) of a black body to its surface area and temperature:

$$L = \sigma A T^4$$

where σ (Stefan-Boltzmann constant) = $5.67 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$, A = surface area (m^2), T = absolute temperature (K), L = luminosity / total power output (W). Note: T must always be in kelvin, not degrees Celsius.

For a **spherical star** of radius r , the surface area is $4\pi r^2$, so:

$$L = 4\pi r^2 \sigma T^4$$

Key Points

- $L \propto T^4$ — doubling the temperature increases luminosity 16-fold.
- $L \propto A$ — a star with four times the surface area has four times the luminosity (at the same temperature).
- A larger, cooler star can have the same luminosity as a smaller, hotter star.
- The law can be used to compare stars or estimate a star's radius from its temperature and luminosity.

Worked Example — Luminosity of the Sun

The Sun has radius $r = 7.0 \times 10^8 \text{ m}$ and surface temperature $T = 5800 \text{ K}$. Calculate the luminosity of the Sun.

Solution:

$$L = 4\pi r^2 \sigma T^4$$

$$L = 4\pi \times (7.0 \times 10^8)^2 \times 5.67 \times 10^{-8} \times (5800)^4$$

$$L = 4\pi \times 4.9 \times 10^{17} \times 5.67 \times 10^{-8} \times 1.13 \times 10^{15}$$

$$L = 3.95 \times 10^{26} \text{ W}$$

This is the accepted value for the Sun's luminosity.

Worked Example — Finding a Star's Radius

A star has the same luminosity as the Sun (3.95×10^{26} W) but a surface temperature of 4100 K. Calculate its radius.

Solution:

From $L = 4\pi r^2 \sigma T^4$:

$$r^2 = L / (4\pi \sigma T^4)$$

$$r^2 = 3.95 \times 10^{26} / (4\pi \times 5.67 \times 10^{-8} \times (4100)^4)$$

$$r^2 = 3.95 \times 10^{26} / (4\pi \times 5.67 \times 10^{-8} \times 2.83 \times 10^{14})$$

$$r^2 = 1.96 \times 10^{19} \text{ m}^2$$

$$r = 1.40 \times 10^9 \text{ m} \approx 2 \times \text{radius of Sun}$$

This is a red giant: cooler and larger than the Sun but with the same luminosity.

Exam Practice Questions

Q1. State the Stefan-Boltzmann law and write down the equation. [2 marks]

Answer: The total power output (luminosity) of a black body is directly proportional to its surface area and to the fourth power of its absolute temperature. $L = \sigma A T^4$ to the power 4, where $\sigma = 5.67 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$.

Q2. Star Sirius has a surface temperature of 12 000 K and a diameter of 2 220 000 km. Calculate its luminosity. [3 marks]

Answer: $r = 1.11 \times 10^9 \text{ m}$. $A = 4\pi r^2 = 4\pi \times (1.11 \times 10^9)^2 = 1.55 \times 10^{19} \text{ m}^2$. $L = \sigma A T^4 = 5.67 \times 10^{-8} \times 1.55 \times 10^{19} \times (12000)^4 = 5.67 \times 10^{-8} \times 1.55 \times 10^{19} \times 2.07 \times 10^{16} = 1.82 \times 10^{28} \text{ W}$ (about 46 times the Sun's luminosity).

9. Wien's Displacement Law

Spec ref: 5.6.163

Statement of Wien's Law

Wien's displacement law states that the **peak wavelength** λ_{max} of radiation emitted by a black body is **inversely proportional to its absolute temperature T**.

$$\lambda_{\text{max}} \times T = 2.898 \times 10^{-3} \text{ m K}$$

where λ_{max} is the peak wavelength (m), T is the absolute temperature (K), and the constant $2.898 \times 10^{-3} \text{ m K}$ is Wien's constant. The unit "m K" means metres $\times$ kelvin (NOT millikelvin).

Rearranging to find temperature from peak wavelength:

$$T = 2.898 \times 10^{-3} \div \lambda_{\text{max}}$$

Physical Meaning

- As temperature **increases**, λ_{max} **decreases** (shifts to shorter wavelengths / higher frequencies).
- Shorter wavelengths = higher energy photons. Hotter objects radiate more energetically.
- The colour of a star (red, yellow, white, blue) is directly related to its surface temperature through Wien's law.
- Wien's law can be used as a **thermometer** for any source of light, including stars billions of light-years away.

Worked Example — Temperature of Betelgeuse

The spectrum of Betelgeuse (in the constellation Orion) has its peak wavelength at $9.35 \times 10^{-7} \text{ m}$. What is the surface temperature of Betelgeuse?

Solution:

$$\lambda_{\text{max}} \times T = 2.898 \times 10^{-3}$$

$$T = 2.898 \times 10^{-3} / \lambda_{\text{max}}$$

$$T = 2.898 \times 10^{-3} / (9.35 \times 10^{-7})$$

$$T = 3100 \text{ K}$$

This low temperature (compared to the Sun's 5800 K) confirms Betelgeuse is a cool red star.

Worked Example — Peak Wavelength of the Sun

The Sun's surface temperature is 5800 K. Calculate the peak wavelength of solar radiation.

Solution:

$$\lambda_{\text{max}} = 2.898 \times 10^{-3} / T$$

$$\lambda_{\text{max}} = 2.898 \times 10^{-3} / 5800$$

$$\lambda_{\text{max}} = 5.00 \times 10^{-7} \text{ m} = 500 \text{ nm}$$

This lies in the green-yellow part of the visible spectrum, consistent with the Sun appearing white-yellow.

Exam Practice Questions

Q1. State Wien's displacement law and write down the equation. [2 marks]

Answer: Wien's displacement law states that the peak wavelength of radiation emitted by a black body is inversely proportional to its absolute temperature. $\lambda_{\text{max}} \times T = 2.898 \times 10^{-3} \text{ m K}$.

Q2. A star of spectral class A has a surface temperature of 9000 K. Calculate its peak wavelength. [2 marks]

Answer: $\lambda_{\text{max}} = 2.898 \times 10^{-3} / 9000 = 3.22 \times 10^{-7} \text{ m} = 322 \text{ nm}$. This is in the ultraviolet region, just outside the visible spectrum.

Q3. Explain how Wien's law can be used to determine the temperature of a star from observations made on Earth. [3 marks]

Answer: By measuring the intensity of light from a star at different wavelengths (its spectrum), we can identify the peak wavelength λ_{max} — the wavelength at which the intensity is greatest. Using Wien's law: $T = 2.898 \times 10^{-3} / \lambda_{\text{max}}$, the surface temperature can be calculated. The star behaves approximately as a black body, so the shape of its spectrum matches the theoretical black body curve for that temperature.

10. Intensity and Luminosity

Spec ref: 5.6.164

Definitions

Luminosity L is the **total power output** of a star — the total energy radiated per second in all directions. Unit: watts (W).

Intensity I is the **power received per unit area** at a given distance from the source. Unit: watts per metre squared (W m^{-2}). Intensity gives the effective brightness as received at a detector.

The Inverse-Square Law for Intensity

Assuming light is emitted equally in all directions, it spreads over the surface of a sphere of radius d . Since the area of a sphere is $4\pi d^2$:

$$I = L \div (4 \pi d^2)$$

where I = intensity (W m^{-2}), L = luminosity (W), d = distance from source (m). Intensity follows the inverse-square law: $I \propto 1/d^2$.

Rearranged to find distance from luminosity and measured intensity:

$$d = \sqrt{[L \div (4 \pi I)]}$$

Worked Example — Solar Intensity at Earth (Solar Constant)

The Sun has luminosity $L = 3.8 \times 10^{26}$ W. The Earth-Sun distance $d = 1.5 \times 10^{11}$ m. Calculate the intensity of sunlight reaching Earth.

Solution:

$$I = L / (4\pi d^2)$$

$$I = 3.8 \times 10^{26} / (4\pi \times (1.5 \times 10^{11})^2)$$

$$I = 3.8 \times 10^{26} / (2.83 \times 10^{23})$$

$$I = 1343 \text{ W m}^{-2} \approx 1350 \text{ W m}^{-2} \text{ (the solar constant)}$$

The slight discrepancy from the accepted 1361 W m^{-2} is due to rounding.

Worked Example — Distance to Betelgeuse

The luminosity of Betelgeuse is $L = 5.4 \times 10^{30}$ W. Its radiant energy intensity measured at Earth is $I = 1.1 \times 10^{-8} \text{ W m}^{-2}$. How far away is Betelgeuse?

Solution:

$$d = \sqrt{L / 4\pi I}$$

$$d = \sqrt{(5.4 \times 10^{30} / (4\pi \times 1.1 \times 10^{-8}))}$$

$$d = \sqrt{(3.90 \times 10^{37})}$$

$$d = 6.2 \times 10^{18} \text{ m} \approx 660 \text{ ly} \approx 200 \text{ pc}$$

Exam Practice Questions

Q1. Define luminosity and intensity, and state their units. [2 marks]

Answer: Luminosity L is the total power output of a star (energy radiated per second in all directions); unit watts (W). Intensity I is the power received per unit area at a given distance from the star; unit W m^{-2} .

Q2. Explain why intensity follows an inverse-square law. [3 marks]

Answer: Light from a star spreads equally in all directions over the surface of a sphere. The surface area of a sphere is $4\pi d^2$, which increases as d^2 . Since the same total power L is spread over this increasing area, $I = L/(4\pi d^2)$, so $I \propto 1/d^2$. Doubling the distance spreads the light over four times the area, reducing intensity by factor 4.

Q3. A star has luminosity $2.5 \times 10^{29} \text{ W}$. An observer on Earth measures its intensity as $8.0 \times 10^{-10} \text{ W m}^{-2}$. Calculate the distance to the star in parsecs. (1 pc = $3.09 \times 10^{16} \text{ m}$) [4 marks]

Answer: $d = \sqrt{(L / 4\pi I)} = \sqrt{(2.5 \times 10^{29} / (4\pi \times 8.0 \times 10^{-10}))} = \sqrt{(2.49 \times 10^{37})} = 1.58 \times 10^{18.5} \text{ m} \dots$
 $d = \sqrt{(2.5 \times 10^{29} / (4 \times 3.14159 \times 8.0 \times 10^{-10}))} = \sqrt{(2.49 \times 10^{37})} = 4.99 \times 10^{18} \text{ m}$. d in pc = $4.99 \times 10^{18} / 3.09 \times 10^{16} = 161 \text{ pc}$.

11. Trigonometric Parallax

Spec ref: 5.6.165

The Parallax Effect

Parallax is the apparent shift in position of a nearby star against the background of more distant stars, as the Earth moves around the Sun. The Earth provides a baseline of 2 AU (diameter of Earth's orbit) for this measurement.

The **parallax angle** θ is half the total apparent angular shift of the star. The larger the parallax angle, the closer the star.

Calculating Distance from Parallax

Using trigonometry for a right-angled triangle (Earth–Sun–star):

$$\tan \theta = r / d \rightarrow d = r / \tan \theta$$

For small angles (which is always the case for stars), $\tan \theta \approx \theta$ (in radians):

$$d = r \div \theta$$

where d = distance to star (m), r = Earth–Sun distance = 1 AU = 1.5×10^{11} m, θ = parallax angle (radians).

Astronomical Distance Units

Unit	Symbol	Definition	Value in metres
Astronomical Unit	AU	Mean Earth–Sun distance	1.5×10^{11} m
Light Year	ly	Distance light travels in 1 year in vacuum	9.46×10^{15} m
Parsec	pc	Distance at which parallax angle = 1 arcsecond	3.09×10^{16} m = 3.26 ly

Shortcut for parsecs: If the parallax angle θ is in **arcseconds**:

$$d \text{ (parsecs)} = 1 \div \theta \text{ (arcseconds)}$$

Worked Example — Distance to Alpha Centauri

Astronomers measure the parallax angle of Alpha Centauri and find that after 6 months the angle changes by 1.52 arcseconds. How far away is Alpha Centauri?

Solution:

Total angular shift = 1.52 arcseconds, so parallax angle $\theta = 1.52/2 = 0.76$ arcseconds.

Using $d = 1/\theta$ (in parsecs): $d = 1/0.76 = 1.32$ pc

Converting: $d = 1.32 \times 3.09 \times 10^{16} = 4.08 \times 10^{16}$ m ≈ 4.3 ly

Alpha Centauri is the nearest star system to the Sun at about 4.2 light years.

Exam Practice Questions

Q1. Explain the method of trigonometric parallax for measuring the distance to a nearby star. [4 marks]

Answer: Observe a nearby star against a background of very distant stars. Record its apparent position. Six months later, when the Earth is on the opposite side of its orbit, record the position again. The star appears to have moved by an angle of 2θ relative to the distant background stars. Using $d = r/\theta$ (or $d = 1/\theta$ in parsecs when θ is in arcseconds), where r is the Earth–Sun distance (1 AU), calculate the distance d .

Q2. Define the parsec. [2 marks]

Answer: One parsec is the distance at which the parallax angle to a star equals exactly 1 arcsecond (where 1 arcsecond = $1/3600$ of a degree). $1 \text{ pc} = 3.09 \times 10^{16} \text{ m} = 3.26 \text{ light years}$.

Q3. Barnard's Star has a parallax angle of 0.545 arcseconds. Calculate its distance in (a) parsecs, (b) metres, (c) light years. [3 marks]

Answer: (a) $d = 1/0.545 = 1.83 \text{ pc}$. (b) $d = 1.83 \times 3.09 \times 10^{16} = 5.66 \times 10^{16} \text{ m}$. (c) $d = 1.83 \times 3.26 = 5.97 \text{ ly}$.

12. Standard Candles

Spec ref: 5.6.166

What are Standard Candles?

Standard candles are astronomical objects whose **luminosity is known independently** of their distance. By measuring the intensity I received at Earth and using the inverse-square law, the distance d can be calculated:

$$d = \sqrt{[L \div (4 \pi I)]}$$

where L = known luminosity (W), I = measured intensity at Earth ($W m^{-2}$), d = distance to the source (m).

Cepheid Variable Stars

Cepheid variables are stars that pulsate in brightness with a regular period. Henrietta Leavitt (1912) discovered that **period is directly proportional to luminosity**: a longer period means greater luminosity.

Procedure for using Cepheids as standard candles:

- Measure the pulsation period of the Cepheid variable.
- Use the period-luminosity relationship to determine the luminosity L .
- Measure the intensity I of light received from the star.
- Calculate the distance: $d = \sqrt{(L / 4\pi I)}$.

Cepheid variables are useful for distances up to about 40 million parsecs (40 Mpc), making them crucial for measuring distances to nearby galaxies.

Type Ia Supernovae

Type Ia supernovae always explode at approximately the same peak luminosity ($\sim 10^{10} \times L_{\text{Sun}}$). They are used as standard candles for the most distant objects — up to billions of parsecs — and were used to discover the accelerating expansion of the Universe.

Main Sequence Fitting

A star's spectral class determines its position on the HR diagram, giving its luminosity. This temperature-luminosity relationship provides another type of standard candle for main sequence stars.

Worked Example — Distance to Betelgeuse (Standard Candle Method)

Betelgeuse has luminosity $L = 5.4 \times 10^{30}$ W. Its measured radiant intensity at Earth is $I = 1.1 \times 10^{-8}$ W m⁻². Find the distance to Betelgeuse.

Solution:

$$d = \sqrt{L / 4\pi I}$$

$$d = \sqrt{(5.4 \times 10^{30} / (4\pi \times 1.1 \times 10^{-8}))}$$

$$d = \sqrt{(3.90 \times 10^{37})} = 6.2 \times 10^{18} \text{ m}$$

$$d \approx 660 \text{ light years} \approx 200 \text{ pc}$$

Exam Practice Questions

Q1. Explain what is meant by a standard candle and how it is used to find the distance to a remote object. [4 marks]

Answer: A standard candle is an astronomical object whose luminosity (total power output) is known independently of its distance. The intensity I of light from the object is measured at Earth. Using $I = L/(4\pi d^2)$, we can rearrange to find $d = \sqrt{L/(4\pi I)}$. Examples: Cepheid variable stars (period-luminosity relation gives L) and Type Ia supernovae (always have the same known peak luminosity).

Q2. Explain the limitation of trigonometric parallax and state why standard candles are needed. [3 marks]

Answer: Trigonometric parallax is only accurate for nearby stars (up to ~100 pc for ground telescopes). For more distant stars and galaxies, the parallax angle becomes too small to measure accurately. Standard candles bypass this limitation by using the known luminosity of objects such as Cepheid variables or Type Ia supernovae, allowing distances to billions of parsecs to be measured.

13. The Hertzsprung-Russell Diagram — Stellar Classification

Spec ref: 5.6.167

Spectral Classification of Stars

Stars are classified into **spectral classes** based on their surface temperature, determined by their absorption line spectra. From hottest to coolest: **O, B, A, F, G, K, M**.

Mnemonic: **Only Brilliant Astronomers Fix Great Kinematic Models**

Class	Colour	Temperature Range (K)	Example
O	Blue	25 000 – 50 000	Zeta Puppis
B	Blue	11 000 – 25 000	Rigel, Spica
A	Blue-White	7 500 – 11 000	Sirius, Vega
F	White	6 000 – 7 500	Canopus
G	Yellow-White	5 000 – 6 000	Sun, Alpha Centauri A
K	Orange	3 500 – 5 000	Arcturus, Aldebaran
M	Red	< 3 500	Betelgeuse, Proxima Centauri

The Hertzsprung-Russell (H-R) Diagram

The **HR diagram** plots stellar **luminosity** (y-axis, increasing upward) against **surface temperature** (x-axis, increasing to the **left**). Equivalently, it can use absolute magnitude and spectral class.

Four main regions:

- **Main sequence:** Diagonal band from upper-left (hot, luminous O-type) to lower-right (cool, dim M-type). Most stars, including the Sun, spend most of their life here. Hydrogen fusion powers main sequence stars.
- **Red giants / Giants:** Upper-right region. Large, cool, luminous stars that have evolved off the main sequence.
- **Supergiants:** Top of the diagram. Most luminous stars, both hot (blue supergiants) and cool (red supergiants).
- **White dwarfs:** Lower-left region. Hot but small (Earth-sized), very dim. Remnants of dead low-mass stars.

Using $L = \sigma \times 4\pi r^2 \times T^4$, a star's radius can be estimated from its position on the HR diagram: a star with the same luminosity as the Sun but lower temperature must be larger. Giants and supergiants are enormous compared to main sequence stars.

Exam Practice Questions

Q1. Sketch the main features of a Hertzsprung-Russell diagram, labelling the main sequence, red giants, white dwarfs, and supergiants. [4 marks]

Answer: The sketch should show: y-axis = luminosity (increasing upward, often in solar units); x-axis = temperature (increasing to the LEFT, or spectral class O-M left to right). Main sequence: diagonal band from top-left to bottom-right. Giants: upper right area (cool, luminous). Supergiants: very top of diagram (most luminous). White dwarfs: lower left area (hot but faint). Accept: absolute magnitude on y-axis, spectral class O B A F G K M on x-axis.

Q2. Explain why the x-axis of an HR diagram has temperature increasing to the left. [2 marks]

Answer: This is a historical convention: the original classification was by spectral type O to M, with O being hottest. When temperature was assigned, the hotter O stars naturally fell on the left. It also means blue stars (hot) are on the left and red stars (cool) are on the right, matching the visual appearance.

Q3. A star has the same luminosity as the Sun but is cooler. What does this tell us about the size of this star? Use the Stefan-Boltzmann law in your answer. [3 marks]

Answer: From $L = 4\pi r^2 \sigma T^4$: if L is the same but T is smaller, then r^2 must be larger to compensate (since $r^2 = L/(4\pi\sigma T^4)$). Therefore the star must have a larger radius — it is a giant or red giant.

14. Stellar Evolution — Life Cycle of Stars

Spec ref: 5.6.168

Overview

The life cycle of a star depends on its **initial mass**. All stars begin in a nebula and end in a compact remnant, but the pathway and final state are determined by mass.

Stage 1: Protostar

A **nebula** (cloud of gas and dust) fragments into clumps under gravity. A clump contracts and rotates faster (conservation of angular momentum), forming a dense, hot **protostar**.

Stage 2: Main Sequence

When core temperature is high enough, **nuclear fusion begins**, fusing hydrogen into helium. The outward radiation pressure balances the inward gravitational collapse (**hydrostatic equilibrium**), making the star stable.

- Higher mass → higher core temperature → faster fusion → **shorter main sequence lifetime**.
- Sun: ~10 billion years on the main sequence. Massive O-type stars: only millions of years.

Low-Mass Pathway (< 3 solar masses)

Red Giant: Hydrogen exhausted in core → core contracts and heats → outer layers expand and cool. Helium fusion begins, making carbon and oxygen.

White Dwarf and Black Dwarf: Outer layers expelled as **planetary nebula**. Core contracts to a **white dwarf** (very hot, very dense, Earth-sized). Eventually cools over billions of years to a theoretically cold **black dwarf** (none have existed long enough to form yet).

High-Mass Pathway (> 3 solar masses)

Red Supergiant: Same process as red giant but larger. Core fuses progressively heavier elements, up to iron. Iron cannot release energy through fusion.

Supernova: When all nuclear fuel is exhausted, the core collapses suddenly and becomes incompressibly rigid. Outer layers rebound in a catastrophic shockwave — a **supernova**. Elements heavier than iron are synthesised and scattered into space.

Remnants after Supernova

Remnant	Mass Condition	Key Properties
Neutron Star	$1.4 - 3 M_{\odot}$ remaining	Gravity forces $p^+ + e^- \rightarrow n$. Density $\sim 10^{17} \text{ kg m}^{-3}$. Radius $\sim 10 \text{ km}$.
Black Hole	$> 3 M_{\odot}$ remaining	Gravity so strong not even light can escape. Defined by Schwarzschild radius: $r_s = 2GM/c^2$.

The HR Diagram and Stellar Evolution

A star's position on the HR diagram changes throughout its life. On the main sequence it is stable. As it evolves off the main sequence it moves right (expanding and cooling to become a giant), then may move left again (as it eventually forms a white dwarf or explodes as a supernova). This evolutionary track is plotted on the HR diagram.

Exam Practice Questions

Q1. Describe the life cycle of a star with a mass similar to the Sun, starting from a nebula. [6 marks]

Answer: Nebula → Protostar (gas and dust contract under gravity, rotate, heat up) → Main sequence star (hydrogen fusion, hydrostatic equilibrium, ~10 billion year lifetime) → Red giant (hydrogen exhausted, core contracts, outer layers expand and cool, helium fusion begins) → Planetary nebula (outer layers expelled) → White dwarf (hot, dense core remains) → Black dwarf (theoretical: cooled white dwarf, takes longer than current age of Universe).

Q2. Explain why a star more massive than the Sun has a shorter main sequence lifetime. [3 marks]

Answer: More massive stars have higher core temperatures and pressures, causing nuclear fusion reactions to occur much faster. They consume their hydrogen fuel more rapidly than lower-mass stars. Although they have more fuel (proportional to mass), they burn it so much faster (rate $\propto M^3$ or higher) that the lifetime is much shorter. An O-type star of 30 solar masses might last only a few million years, compared to 10 billion years for the Sun.

Q3. Explain what happens when the core of a massive star collapses at the end of its life. [4 marks]

Answer: When all nuclear fuel is exhausted, fusion stops and the outward radiation pressure disappears. Gravity causes the core to collapse very rapidly. The core becomes incompressibly rigid when matter cannot be compressed further. The outer layers that were still falling inward rebound off this rigid core in a powerful shockwave. This expels the outer layers in a supernova explosion. As the shockwave passes through the surrounding material, elements heavier than iron are synthesised. The remaining core becomes either a neutron star or black hole, depending on its mass.

15. The Doppler Effect and Cosmological Redshift

Spec ref: 5.6.169

The Doppler Effect

The **Doppler effect** occurs when a wave source moves relative to an observer, causing the observed wavelength (and frequency) to differ from the emitted values.

- Source **moving away**: wavefronts are stretched → longer wavelength → lower frequency → **redshift** (for light).
- Source **moving towards**: wavefronts are compressed → shorter wavelength → higher frequency → **blueshift** (for light).

When astronomers examined light from distant galaxies in the 1920s, they found all spectral absorption lines shifted towards the red end — the galaxies are moving away from us.

Quantifying Redshift

The **redshift parameter** z quantifies how much a spectrum has shifted:

$$z = \Delta\lambda / \lambda \approx \Delta f / f \approx v / c$$

where $\Delta\lambda$ = change in wavelength (observed – emitted), λ = emitted wavelength, Δf = change in frequency, f = emitted frequency, v = recession velocity (m/s), c = speed of light = 3.0×10^8 m/s. This approximation is valid for $v \ll c$.

For large z (very distant galaxies), relativistic corrections are needed, but for most exam purposes the approximation $z = v/c$ is sufficient.

Worked Example — Speed of a Star from Spectral Shift

In a laboratory, the hydrogen alpha line has wavelength $\lambda = 656.285$ nm. In the spectrum of a nearby star, the same line appears at 656.315 nm. How fast is the star moving, and in which direction?

Solution:

The wavelength is longer than expected, so the star is moving away (redshift).

$$\Delta\lambda = 656.315 - 656.285 = 0.030 \text{ nm}$$

$$v = c \times \Delta\lambda / \lambda = 3.0 \times 10^8 \times (0.030 / 656.285)$$

$$v = 3.0 \times 10^8 \times 4.57 \times 10^{-5}$$

$$v = 1.37 \times 10^4 \text{ m/s} = 13.7 \text{ km/s away from Earth}$$

Worked Example — Distance from Redshift

A supernova is observed with redshift $z = 0.45$. How far away is it? ($H_0 = 70.9 \text{ km s}^{-1} \text{ Mpc}^{-1}$)

Solution:

$$v = z \times c = 0.45 \times 3.0 \times 10^8 = 1.35 \times 10^8 \text{ m/s} = 1.35 \times 10^5 \text{ km/s}$$

$$\text{Using Hubble's law: } d = v / H_0 = 1.35 \times 10^5 / 70.9 = 1900 \text{ Mpc}$$

Exam Practice Questions

Q1. Explain what is meant by redshift and how it arises. [3 marks]

Answer: Redshift is the increase in wavelength (and decrease in frequency) of electromagnetic radiation from a source that is moving away from the observer. As the source moves away, successive wavefronts are emitted from a greater distance, increasing the spacing between them (the wavelength). This is the Doppler effect applied to light.

Q2. The galaxy NGC 7320C has a redshift $z = 0.02$. (a) At what wavelength would you find the hydrogen alpha absorption line (emitted at 656.3 nm)? (b) Calculate the distance to the galaxy using $H_0 = 70.9 \text{ km s}^{-1} \text{ Mpc}^{-1}$. [4 marks]

Answer: (a) $z = \Delta\lambda/\lambda \rightarrow \Delta\lambda = z \times \lambda = 0.02 \times 656.3 = 13.1 \text{ nm}$. Observed wavelength = $656.3 + 13.1 = 669.4 \text{ nm}$. (b) $v = z \times c = 0.02 \times 3.0 \times 10^8 = 6.0 \times 10^6 \text{ m/s} = 6000 \text{ km/s}$. $d = v/H_0 = 6000/70.9 = 84.6 \text{ Mpc}$.

16. Hubble's Law and the Expanding Universe

Spec ref: 5.6.170

Hubble's Discovery

In 1929, Edwin Hubble found that the recession velocity of a galaxy is **directly proportional to its distance** from us. He used redshift measurements (recession velocities) and standard candle distances (Cepheid variables) for many galaxies and found a linear relationship:

$$v = H_0 d$$

where v = recession velocity (km s^{-1}), d = distance to the galaxy (Mpc), H_0 = Hubble constant $\approx 70.9 \text{ km s}^{-1} \text{ Mpc}^{-1} \approx 2.29 \times 10^{-18} \text{ s}^{-1}$ (in SI units).

This finding implies the Universe is **expanding**: all galaxies are moving apart from each other. The further a galaxy is, the faster it recedes.

Age of the Universe

If the Universe has expanded at a constant rate since the Big Bang:

$$T_0 = 1 \div H_0$$

T_0 = age of the Universe; H_0 must be in SI units (s^{-1}). $1 \text{ Mpc} = 3.09 \times 10^{22} \text{ m}$, so $H_0 = 70.900 \text{ m/s} / 3.09 \times 10^{22} \text{ m} = 2.29 \times 10^{-18} \text{ s}^{-1}$.

$$T_0 = 1 / 2.29 \times 10^{-18} = 4.37 \times 10^{17} \text{ s} \approx \mathbf{13.8 \text{ billion years}}.$$

Uncertainty in the Hubble Constant

Different measurement methods give H_0 values ranging from about 67 to 73 $\text{km s}^{-1} \text{ Mpc}^{-1}$. This "Hubble tension" corresponds to an age range of roughly 13.3–14.6 billion years. The exact value depends on how distances are calibrated — this remains an active area of research.

Worked Example — Galaxy's Recession Velocity

A supernova appears in the night sky with redshift $z = 0.45$. How far away is it? $H_0 = 70.9 \text{ km s}^{-1} \text{ Mpc}^{-1}$.

Solution:

From $z \approx v/c$: $v = z \times c = 0.45 \times 3.0 \times 10^8 \text{ m/s} = 1.35 \times 10^8 \text{ m/s} = 1.35 \times 10^5 \text{ km/s}$

Using Hubble's law: $d = v / H_0 = 1.35 \times 10^5 / 70.9 = 1904 \text{ Mpc} \approx 1900 \text{ Mpc}$

Worked Example — Age of the Universe

Calculate the age of the Universe if $H_0 = 70.9 \text{ km s}^{-1} \text{ Mpc}^{-1}$.

Solution:

Convert H_0 to SI: $H_0 = 70\,900 \text{ m s}^{-1} \text{ Mpc}^{-1}$

$$1 \text{ Mpc} = 3.09 \times 10^{22} \text{ m}$$

$$H_0 = 70\,900 / 3.09 \times 10^{22} = 2.29 \times 10^{-18} \text{ s}^{-1}$$

$$T_0 = 1 / H_0 = 1 / 2.29 \times 10^{-18} = 4.37 \times 10^{17} \text{ s}$$

$$\approx 4.37 \times 10^{17} / (3.15 \times 10^7) = 13.8 \times 10^9 \text{ years} = 13.8 \text{ billion years}$$

Exam Practice Questions

Q1. State Hubble's law and explain what it tells us about the Universe. [3 marks]

Answer: Hubble's law: $v = H_0 d$, where v is the recession velocity of a galaxy, d is its distance, and H_0 is the Hubble constant. It tells us that the recession velocity is directly proportional to distance. All galaxies are moving away from us (and from each other), meaning the Universe is expanding. Extrapolating backward in time, all matter was once concentrated at a single point — the Big Bang.

Q2. Explain why the value of the Hubble constant affects our estimate of the age of the Universe. [3 marks]

Answer: The age $T_0 = 1/H_0$. A larger H_0 means a faster rate of expansion, so the Universe would have taken less time to reach its current size — younger age. A smaller H_0 implies slower expansion and a greater age. Uncertainty in measuring H_0 (due to uncertainties in measuring stellar distances) directly translates to uncertainty in the age of the Universe.

Q3. A galaxy has a recession velocity of 8400 km/s. Using $H_0 = 70 \text{ km s}^{-1} \text{ Mpc}^{-1}$, calculate the distance to the galaxy in Mpc and in metres. [3 marks]

Answer: $d = v / H_0 = 8400 / 70 = 120 \text{ Mpc}$. d in metres = $120 \times 3.09 \times 10^{22} = 3.71 \times 10^{24} \text{ m}$.

17. The Fate of the Universe: Dark Matter and Dark Energy

Spec ref: 5.6.171

Critical Density and the Fate of the Universe

The future expansion of the Universe depends on the total mass-energy density compared to the **critical density**:

Density vs Critical	Type of Universe	Fate
$\rho > \rho_{\text{critical}}$	Closed (spherical)	Expansion slows, stops, then collapses: Big Crunch
$\rho = \rho_{\text{critical}}$	Flat	Expansion slows but never quite stops: infinite expansion
$\rho < \rho_{\text{critical}}$	Open (hyperbolic)	Expansion continues forever, getting faster: heat death

Dark Matter

In the 1960s–70s, astronomer **Vera Rubin** measured the orbital speeds of stars in galaxies. Unexpectedly, stars far from the galactic centre moved almost as fast as those near the centre.

According to Newton's law of gravitation, if all mass is concentrated at the centre, orbital speed should decrease with distance ($v^2 = GM/r \rightarrow v \propto 1/\sqrt{r}$). But this was not observed. The implication: there is a large amount of **invisible ("dark") matter** distributed through and around the galaxy.

- Dark matter accounts for about **26.8%** of the Universe's total energy content.
- Its nature remains unknown; candidates include WIMPs (Weakly Interacting Massive Particles) and axions.
- Dark matter cannot be seen directly but can be inferred from gravitational effects.

Gravitational Lensing

Einstein's general relativity predicts that massive objects warp spacetime, bending light paths. This **gravitational lensing** produces distorted or multiple images of background galaxies around massive foreground galaxy clusters.

Gravitational lensing around clusters shows **more bending than visible matter can explain**, providing independent evidence for dark matter. NASA's Hubble Space Telescope has mapped dark matter distributions using lensing.

Dark Energy

In the late 1990s, measurements of **Type Ia supernovae** showed that distant supernovae are fainter than expected — they are further away than Hubble's law with a decelerating universe would predict. This means the expansion of the Universe is **accelerating**.

This acceleration is attributed to **dark energy** — a mysterious form of energy that permeates all space and counteracts gravity, driving the expansion faster.

- Dark energy accounts for about **68.3%** of the Universe's total energy content.
- Ordinary (baryonic) matter is only about **4.9%** of total energy.
- The nature of dark energy is one of the greatest unsolved problems in physics.

Summary of Universal Composition

Component	Percentage	Nature
Ordinary (baryonic) matter	~4.9%	Stars, gas, planets, us — everything visible
Dark matter	~26.8%	Invisible; inferred from gravity; nature unknown
Dark energy	~68.3%	Drives accelerating expansion; nature unknown

Exam Practice Questions

Q1. Describe the evidence from galaxy rotation curves for the existence of dark matter. [4 marks]

Answer: Vera Rubin observed that stars in spiral galaxies at large distances from the galactic centre orbit at approximately the same speed as stars near the centre. Newton's law of gravitation predicts that if most mass is at the centre, orbital speed should decrease with distance ($v \propto 1/\sqrt{r}$). The observed flat rotation curve cannot be explained by visible matter alone. The discrepancy suggests there is a large amount of additional invisible (dark) matter distributed throughout the galaxy, providing extra gravitational force on the outer stars.

Q2. Explain what is meant by gravitational lensing and how it provides evidence for dark matter. [3 marks]

Answer: Gravitational lensing is the bending of light paths by massive objects, predicted by Einstein's general relativity (mass curves spacetime). Light from distant galaxies is bent around massive foreground galaxy clusters, producing distorted or multiple images. The amount of bending observed is greater than can be explained by the visible mass of the cluster, implying the presence of additional dark matter.

Q3. Explain why the discovery of the accelerating expansion of the Universe was surprising, and describe what is thought to be responsible. [4 marks]

Answer: Before the late 1990s, it was expected that the expansion of the Universe would be slowing down due to the attractive gravitational force between all matter. Type Ia supernovae observations showed that distant supernovae were dimmer (further away) than expected, implying the expansion is speeding up rather than slowing. This acceleration is attributed to dark energy — a mysterious energy that permeates all of space and acts repulsively on large scales, counteracting gravity. Dark energy is estimated to make up about 68% of the Universe's total energy content.